

Coupled FEM/BEM: Open-Domain Conditions in 2D Electrostatics

Computational Electromagnetics

Matteo Morana

Master in Mathematical Engineering
Politecnico di Milano

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The Problem: Parallel-Plate Capacitor in 2D

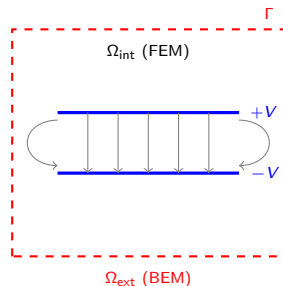
Solve Laplace's equation in the whole plane:

$$\Delta u = 0 \quad \text{in } \mathbb{R}^2 \setminus \{\text{plates}\}$$

with Dirichlet conditions on the plates:

$$u = +V \text{ (top)}, \quad u = -V \text{ (bottom)}$$

Goal: compute the electric potential *both inside and outside* the capacitor in a real case scenario.



Why Not Pure FEM? [fem·bem·text]

Pure FEM approach

Truncate the domain; impose $u = 0$ on the artificial boundary Γ .

Drawbacks:

- Requires a **very large domain** to minimize the truncation error (Γ must be placed at $\sim 50 L_y$ from the plates).
- Leads to **large meshes** and long computation times.
- Cannot represent the solution **up to infinity**: introduces an unavoidable approximation.

Key idea

Use **FEM** inside a small domain and **BEM** to handle the exterior exactly — including infinity.

FEM Weak Formulation

Starting from the Laplace problem on a bounded domain Ω :

$$\Delta u = 0 \quad \Rightarrow \quad \int_{\Omega} \nabla u \cdot \nabla v \, dx - \underbrace{\int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds}_{\text{boundary term}} = 0 \quad \forall v \in H^1(\Omega)$$

After discretisation with basis $\{v_i\}_{i=1}^{N_h}$, writing $u \approx \sum_j u_j v_j$ and $\frac{\partial u}{\partial n} \approx \sum_j \frac{\partial u_j}{\partial n} v_j$:

$$\mathbf{S} \mathbf{u} - \mathbf{T} \frac{\partial \mathbf{u}}{\partial n} = \mathbf{0}$$

where $S_{ij} = \int_{\Omega} \nabla v_i \cdot \nabla v_j \, dx$ and $T_{ij} = \int_{\partial\Omega} v_i v_j \, ds$.

Variable Count — The Key Issue

How many unknowns?

The system provides N_h equations. The unknowns are:

- $u_j, j = 1, \dots, N_h$ nodal potential values
- $\partial u_j / \partial n, j = 1, \dots, N_h$ normal-derivative coefficients

N_h equations vs $2N_h$ unknowns $\Rightarrow$ **under-determined!**

Rescue: Lagrange nodal basis functions satisfy $v_i(x_j) = \delta_{ij}$, so

$v_i|_{\partial\Omega} = 0$ for every *interior* node $i \implies T_{ij} = 0$ if i or j is interior.

Variable reduction

- Normal-derivative unknowns at *interior* nodes drop out of the system.
- System: N_h equations, $N_h + N_h^\partial$ unknowns. Here BEM provide N_h^∂ equations.

FEM Block Structure

Partitioning nodes into **interior** (I) and **boundary** (B):

$$\underbrace{\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} & S_{BB} \end{bmatrix}}_{\text{FEM stiffness}} \begin{bmatrix} u_I \\ u_B \end{bmatrix} - \begin{bmatrix} 0 \\ T_{BB} \frac{\partial u_B}{\partial n} \end{bmatrix} = 0$$

- Row I : closed — $\partial u / \partial n$ does not appear.
- Row B : **open** — involves the unknown $\partial u_B / \partial n$ (will be supplied by BEM).

BEM — Boundary Integral Equation [bem·code]

Key idea: reformulate the Laplace problem on $\partial\Omega$ only, using the fundamental solution $G(x, y) = \frac{1}{2\pi} \log |x - y|$ in 2D.

Green's third identity $\Rightarrow$ **Boundary Integral Equation (BIE):**

$$\frac{1}{2} u(y) = \underbrace{\int_{\partial\Omega} G(x, y) \frac{\partial u}{\partial n}(x) ds_x}_{\mathcal{V}(\partial u / \partial n)(y)} - \underbrace{\int_{\partial\Omega} \frac{\partial G}{\partial n_x}(x, y) u(x) ds_x}_{\mathcal{K}(u)(y)} \quad \forall y \in \partial\Omega$$

After collocation discretizations with boundary basis $\{\psi_j\}$:

$$H \mathbf{u}_B = G \frac{\partial \mathbf{u}_B}{\partial n} \quad H = \frac{1}{2} M + K, \quad G \text{ symmetric}$$

BEM provides **exactly** N_h^∂ equations relating $\mathbf{u}_B$ and $\partial \mathbf{u}_B / \partial n$ — the missing relations from FEM.

Coupled System and Plates Conditions

Replace the open FEM boundary rows with the BEM equations:

$$\begin{bmatrix} S_{FF} & S_{FD} & S_{FB} \\ S_{DF} & S_{DD} & S_{DB} \\ S_{BF} & S_{BD} & S_{BB} \end{bmatrix} \begin{bmatrix} u_F \\ u_D \\ u_B \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ -T_{BB} \frac{\partial u_B}{\partial n} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- **Row 1 (FEM, free):** governs u at unconstrained interior nodes.
- **Row 2 (FEM, Dirichlet):** enforces the prescribed potential on the plates.
- **Row 3 (FEM–BEM, boundary):** couples the FEM domain to the exterior via $T_{BB} G^{-1} H$.

Elimination and Reduced System

Assume G invertible:

$$\frac{\partial u_B}{\partial n} = -G^{-1}H u_B$$

Substitute into the FEM boundary row $\Rightarrow$ **reduced system of size N_h** :

$$\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} & S_{BB} + \underbrace{T_{BB} G^{-1} H}_{\text{BEM correction (dense)}} \end{bmatrix} \begin{bmatrix} u_I \\ u_B \end{bmatrix} = - \begin{bmatrix} S_{ID} \\ S_{BD} \end{bmatrix} u_D$$

- Same size as a standard FEM system ($N_h \times N_h$).
- The BEM term $T_{BB}G^{-1}H$ is **dense** but restricted to boundary DOFs.
- In MATLAB: $G^{-1}H$ computed via the backslash operator.
- After solving: recover $\partial u_B/\partial n$ and the exterior potential via the BEM representation formula.

Setup

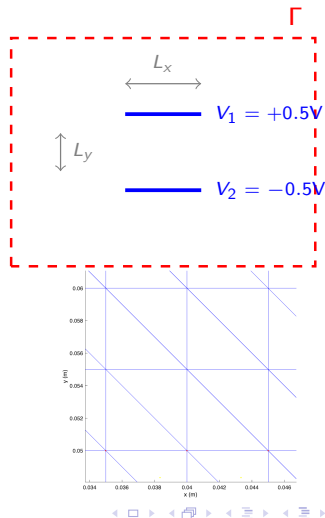
Geometry:

- $L_x = 0.05$ m, $L_y = 0.05$ m ($L_x/L_y = 1$)
- $V_1 = 0.5$ V (top), $V_2 = -0.5$ V (bottom)
- Γ placed at $3 L_y$ margin on all sides
- $\varepsilon_r = 1$

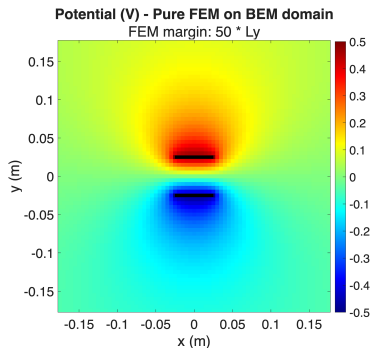
Meshes:

- Coarse: $\delta = 0.005$ m ≈ 5000 nodes

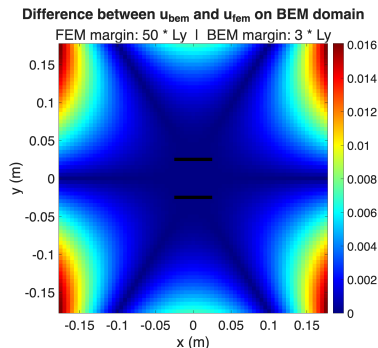
Reference: The reference is a pure FEM method with a Dirichlet condition imposed on Γ placed at $50L_y$ ($\approx 1 \times 10^6$ nodes)



FEM-BEM vs. Pure FEM



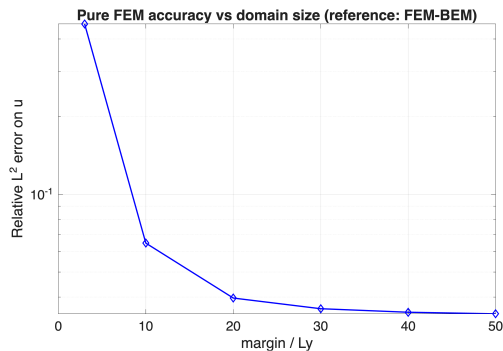
Pure FEM ($50 L_y$ domain) — reference solution



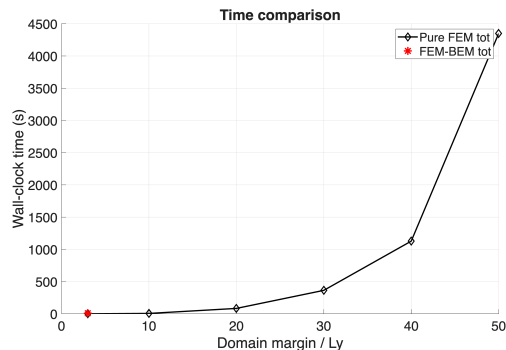
Pointwise difference: FEM vs. FEM-BEM on common domain

$$\varepsilon_{L^2} = \sqrt{\frac{\sum_{i \in \mathcal{N}_c} (u_i^{\text{FEM}} - u_i^{\text{FEM-BEM}})^2}{\sum_{i \in \mathcal{N}_c} (u_i^{\text{FEM}})^2}} \times 100\% = \mathbf{3.44\%}$$

FEM-BEM vs. Pure FEM — Accuracy & Speed



L^2 error vs. FEM domain size

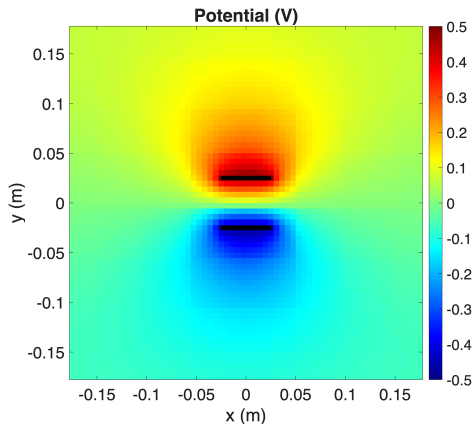


Computational time comparison

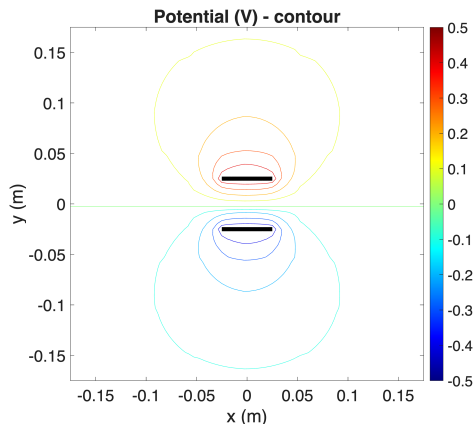
Efficiency gain *on my computer*

- FEM-BEM ($3 L_y$ domain): **8.90 s**
- Pure FEM ($50 L_y$ domain): **4350 s** (≈ 72 min)

Electric Potential Distribution FEM-BEM



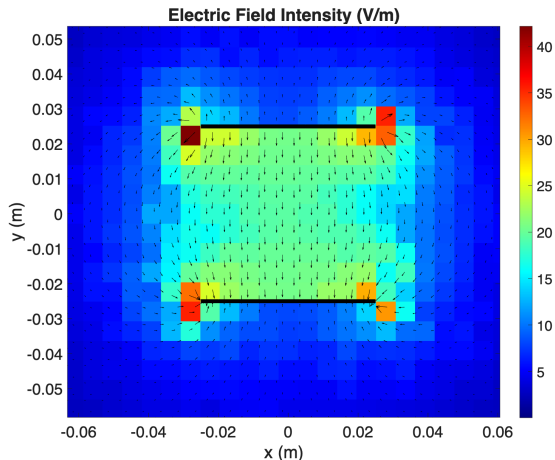
Potential map ($\delta = 0.005$ m)



Equipotential lines

- Uniform field between plates; smooth fringing at edges.

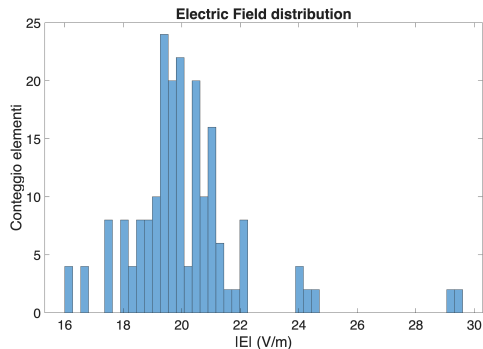
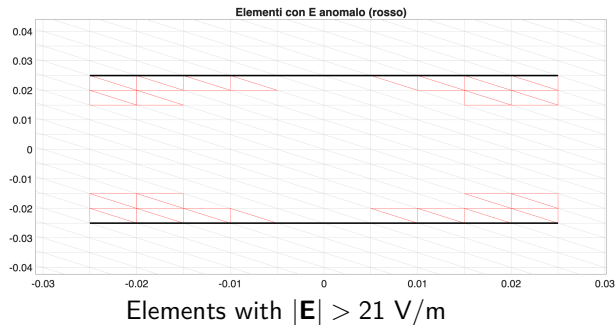
Electric Field FEM-BEM



$\delta = 0.005$ m (coarse)

- Central field $\approx (V_1 - V_2)/L_y = 20$ V/m as expected.
- Fine mesh resolves singularity at plate corners (physically expected).
- Coarse mesh smears the edge singularity.

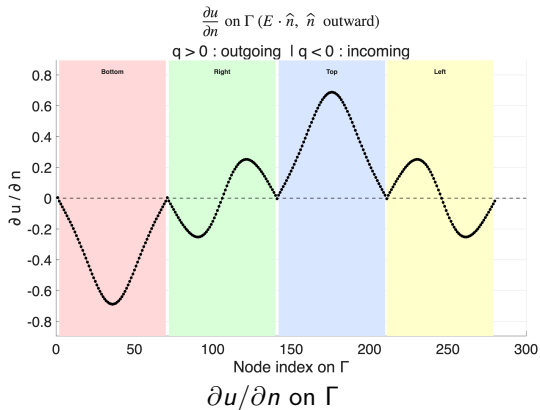
Electric Field — Element Analysis (FEM-BEM)



Distribution of $|\mathbf{E}|$ per element

- Anomalous elements localised at plate **corners** only.
- This is a well-known physical singularity of infinitely thin conductors — **not** a numerical artefact.
- The majority of elements have $|\mathbf{E}| \approx 20 \text{ V/m}$.

Flux Through Γ (FEM-BEM)



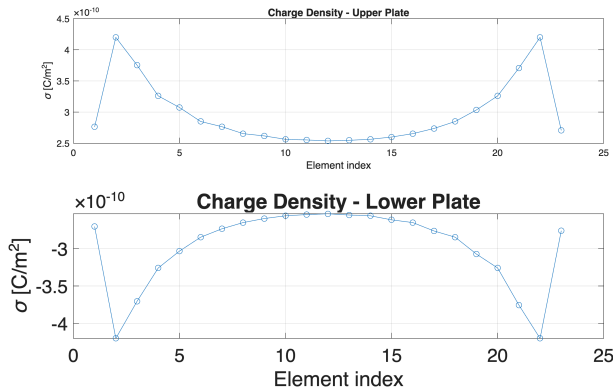
Physical expectation:

- Net flux must vanish (equal and opposite plate charges).

$$Q_{tot} = \oint_{\Gamma} \frac{\partial u}{\partial n} ds \approx -1.79 \times 10^{-13} \approx 0$$

Numerically consistent with zero — coupling condition verified.

Surface Charge Density and Capacitance



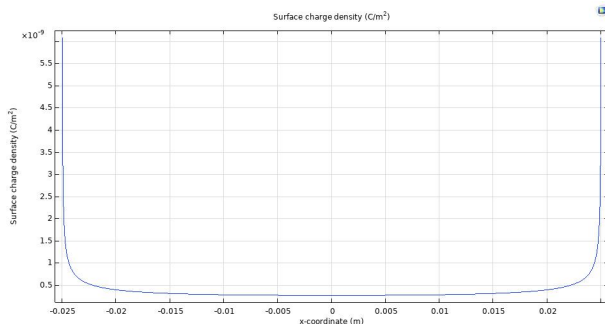
Surface charge density on the plates FEM-BEM

Charge method:

$$C = \frac{|Q|}{V_1 - V_2}$$

- $C_{\text{FEM-BEM}} = 1.7110 \times 10^{-11} \text{ F/m}$
- $C_{\text{COMSOL}} = 1.8491 \times 10^{-11} \text{ F/m}$
- $C_{\text{err}} \approx 7\%$

COMSOL charge density and Energy Capacitance



Surface charge density on the plates COMSOL

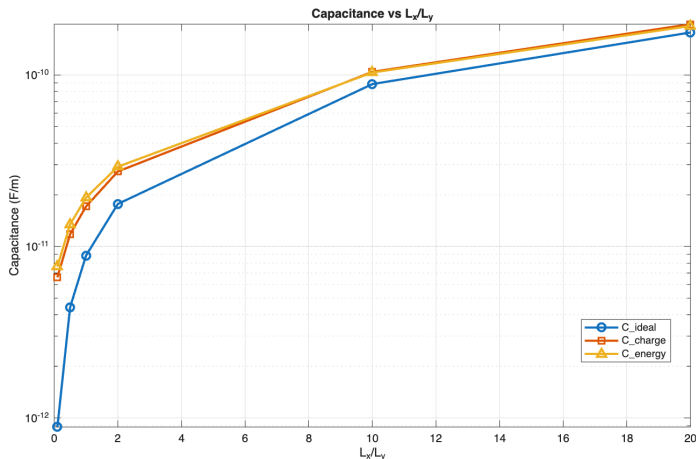
Energy method:

$$W = \frac{1}{2} \int_{\Omega} \varepsilon |\nabla u|^2 d\Omega, \quad (1)$$

$$C = \frac{2W}{\Delta V^2} \quad (2)$$

- $C_{\text{FEM-BEM}} = 1.9284 \times 10^{-11} \text{ F/m}$
- $C_{\text{COMSOL}} = 1.8728 \times 10^{-11} \text{ F/m}$
- $C_{\text{err}} \approx 2.9\%$

Capacitance different L_y , fixed L_x



The capacitance tends toward its ideal value when L_y becomes much larger than L_x , since the shape tends to be ideal.

Conclusions

- A **FEM-BEM coupled solver** for 2D electrostatics has been implemented in MATLAB, targeting the parallel-plate capacitor problem.
- The coupling exploits:
 - ① FEM's accuracy in the bounded interior region.
 - ② BEM's ability to represent the exterior field **up to infinity**.
- Validation confirms:
 - Correct potential and field distributions.
 - Capacitance within 0.69% of the reference FEM solution.
 - Gauss's law satisfied numerically.
- **100× speed-up** over a comparable pure FEM simulation.

Simulation Parameters

Physical Domain

Lx, Ly	Dimensions of the rectangular domain [m]
delx, dely	Spatial discretization step (FEM grid) [m]

Medium Properties

e0	Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
er	Relative permittivity of the medium ϵ_r

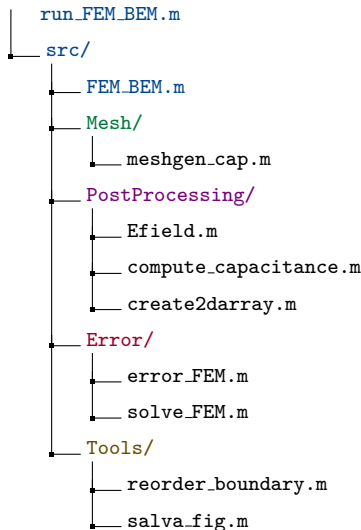
FEM-BEM Coupling

margin_BEM_coeff	Coefficient defining the margin relative to L_y : $\text{margin} = \alpha \cdot L_y$
margini_list	List of BEM margins to test (e.g. error analysis)

Boundary Conditions

V1	Potential applied to conductor 1 [V]
V2	Potential applied to conductor 2 [V]

Code Structure – Overview



Entry point & control

`run_FEM_BEM.m` — sets parameters and calls `FEM_BEM`

`FEM_BEM.m` — main function: orchestrates mesh, assembly, solve and post-processing

Mesh

`meshgen_cap.m` — generates the mesh for the capacitor problem

Post-processing

`Efield.m` — computes the electric field $\mathbf{E} = -\nabla V$

`compute_capacitance.m` — computes capacitance via the energy method

Error Analysis

`solve_FEM.m` — pure FEM solution

`error_FEM.m` — comparison between FEM and FEM–BEM

Utilities

`reorder_boundary.m` — reorders boundary nodes counter-clockwise

`salva_fig.m` — saves figures in the required formats

`create2darray.m` — reconstructs the 2D grid for visualization

Code Structure – src/Assembly

Assembly/

- assemble_S.m
- assemble_T.m
- assemble_G_H.m
- BEM_G_H/
 - findG.m
 - findG_lin.s.m
 - findH.m
 - intG.m
 - intH.m
 - shape_fun.m
 - find_N.m
 - find_R.m
 - abs_J.m

Global assembly functions

`assemble_S.m` — assembles the FEM stiffness matrix

`assemble_T.m` — assembles the FEM–BEM coupling matrix

`assemble_G_H.m` — loops over boundary elements and assembles **G**, **H**, calling the required subroutines

BEM subroutines – boundary integrals

`findG.m` — element-level submatrix G

`findH.m` — element-level submatrix H

`findG_lin.s.m` — version of `findG.m` with singularity treatment

`intG.m` — integrates the kernel $\ln(r)$

`intH.m` — integrates the kernel $1/r$

`shape_fun.m` — evaluates BEM shape functions on the element

`find_N.m` — outward normals on the boundary element

`find_R.m` — distance vector r between field and source point

`abs_J.m` — computes the boundary element Jacobian

References I

The code is available here:

https://github.com/Michelino21/PROJECT_FEM_BEM_FINAL.git